

UNIT 4: CHEMICAL SYSTEMS AND EQUILIBRIUM

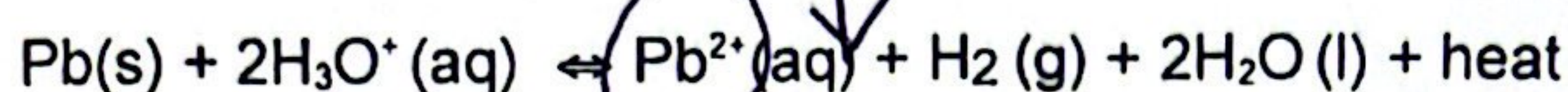
Knowledge	Thinking/ Inquiry	Communication	Application
/15	/18	/8	/9

SHORT ANSWER QUESTIONS

Answer the following questions in the space provided.

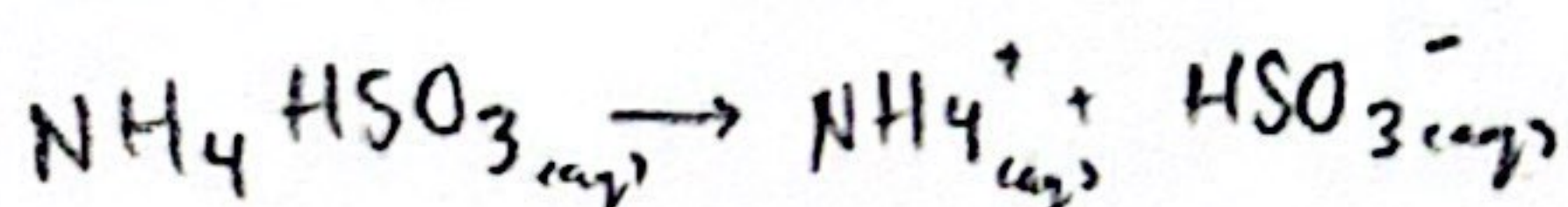
THINKING/INQUIRY (18 MARKS)

1. Which direction will the following equilibrium shift when exposed to the changes below? (3 marks)



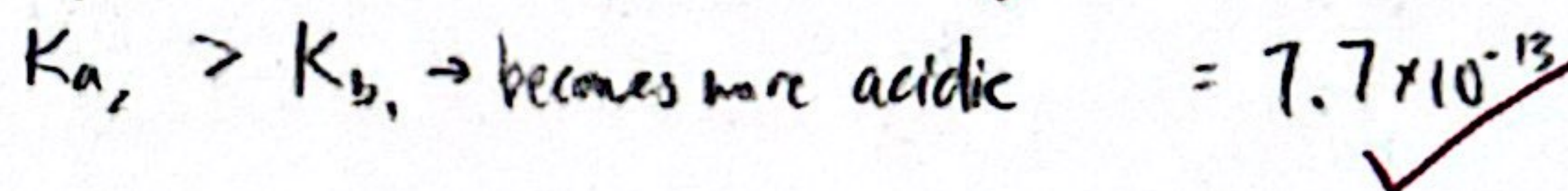
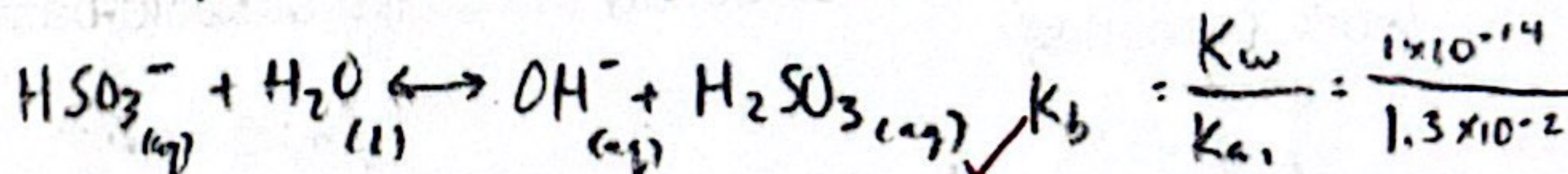
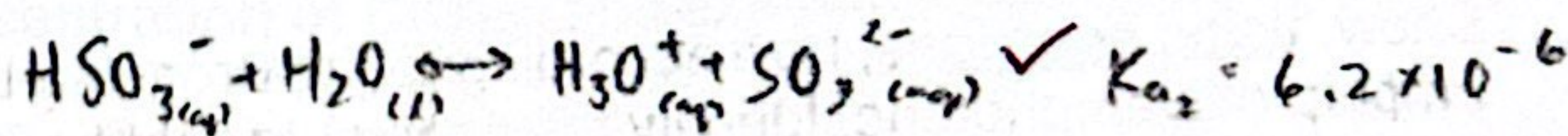
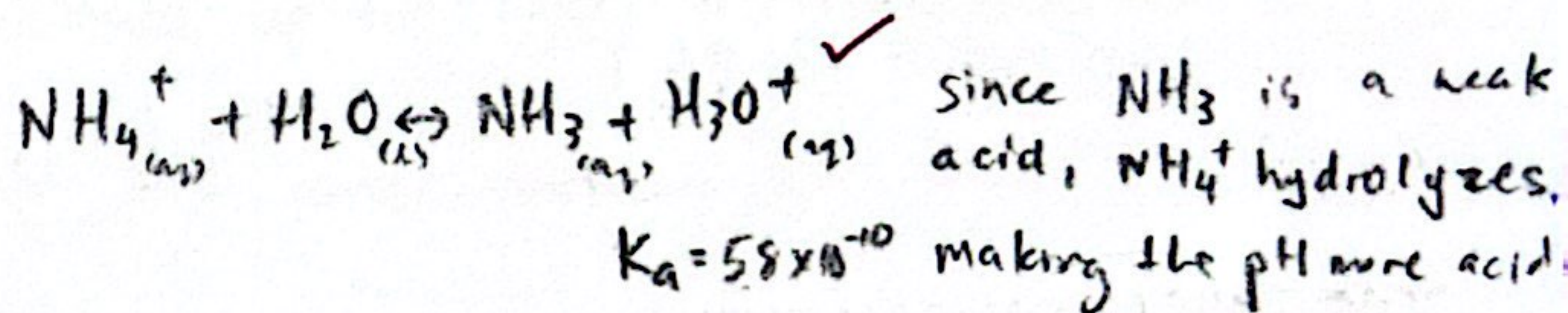
Stress	Direction of Shift
addition of sodium phosphate Na_3PO_4 PO_4^{3-} hydrolyses	left X (other test common ion effect is considered first) $\text{Pb}_3(\text{PO}_4)_2(\text{s})$
addition of OH^- ions $\text{H}_3\text{O}^+ + \text{OH}^- \rightarrow 2\text{H}_2\text{O}$	left ✓
addition of Xe to the container	no change ✓
removal of H_2	right ✓
decreasing the temperature of the reaction	right ✓
decreasing the volume of the container	left ✓

2. Determine whether a solution of ammonium hydrogen sulfite (
- NH_4HSO_3
-) is acidic, basic, or neutral. Show all of your work! (4 marks)

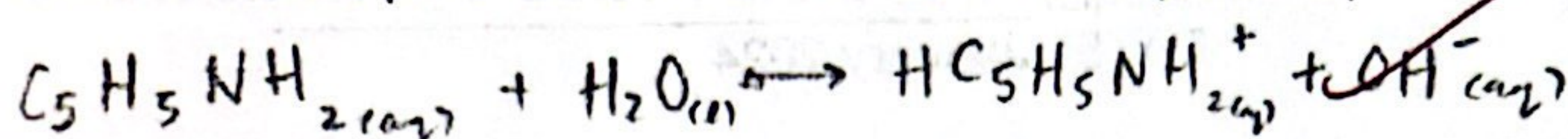


(and a weak conj. acid/base)
 HSO_3^- is amphoteric. The K_a of the equation that produces H_3O^+ ions is greater than the K_b of the reaction that produces OH^- , therefore making the solution more acidic.

The solution will be acidic.

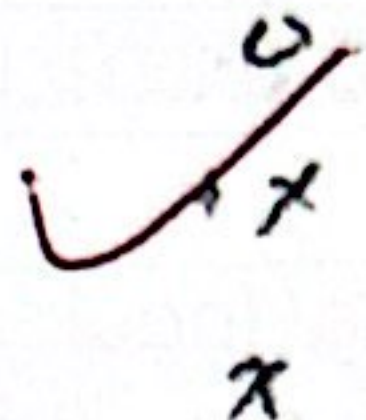


3. Calculate the pH of a 0.80 M solution of aniline. (5 marks)



$$K_b \text{ of aniline} = 4.1 \times 10^{-10}$$

$$\begin{array}{l} I \quad 0.80 \\ C \quad -x \\ E \quad 0.80 - x \end{array}$$



$$K_b = \frac{[HC_5H_5NH_2^+][OH^-]}{[C_5H_5NH_2]} = \frac{x^2}{0.80 - x}$$

$$4.1 \times 10^{-10} = \frac{x^2}{0.80}$$

$$\frac{0.80}{4.1 \times 10^{-10}} = 195121951$$

which is > 100
100 rule applies

$$x = \sqrt{(4.1 \times 10^{-10})(0.80)}$$

$$= 1.811 \times 10^{-5} M = [OH^-]$$

$$pOH = -\log(1.811 \times 10^{-5}) = 4.74$$

$$pH = 14 - 4.74 = 9.26$$

4. What is the pH of a solution when 40.00 mL of 1.20 mol/L NaOH is titrated with 15.00 mL of 0.50 mol/L HBr(aq)? (6 marks)

$$40.00 \text{ mL} \times \frac{1.20 \text{ mol}}{1000 \text{ mL}} = 0.04800 \text{ mol}$$

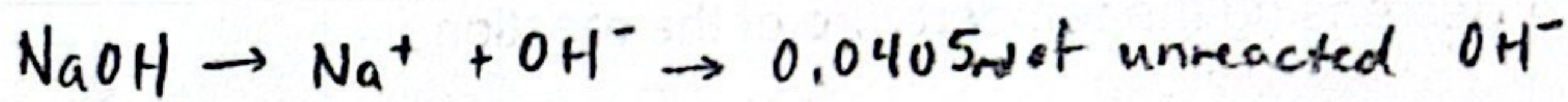
$$15.00 \text{ mL} \times \frac{0.50 \text{ mol}}{1000 \text{ mL}} = 0.007500 \text{ mol}$$

$$n_{NaOH} = (1.20 \text{ mol/L})(0.04000 \text{ L}) = 0.04800 \text{ mol}$$

$$n_{NaOH} - n_{HBr} = 0.048 - 0.0075$$

$$= 0.0405 \text{ mol of NaOH unreacted}$$

$$n_{HBr} = (0.50 \text{ mol/L})(0.01500 \text{ L}) = 0.0075 \text{ mol}$$



$$[OH^-] = 0.0405 \text{ mol} / (0.04000 \text{ L} + 0.01500 \text{ L}) = 0.7364 \text{ mol/L}$$

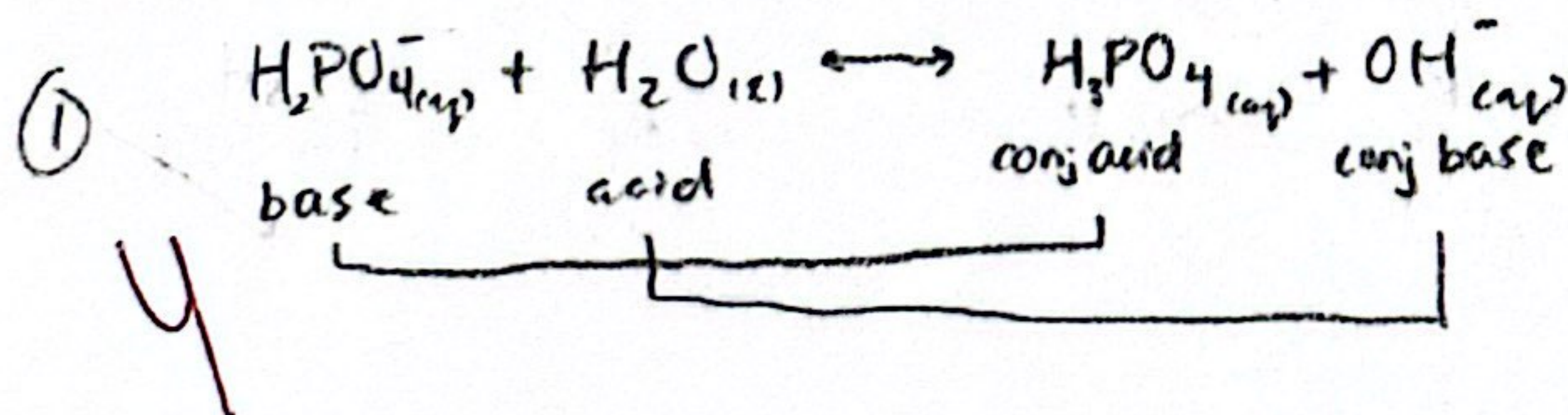
$$pOH = -\log(0.7364) = 0.133$$

$$pH = 14 - 0.133 = 13.867$$

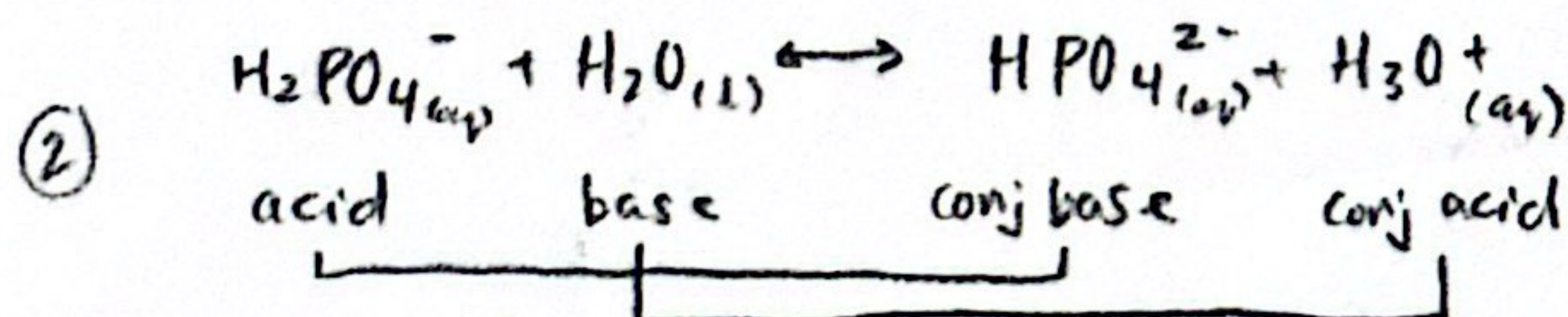
The pH is 13.87

COMMUNICATION (8 MARKS)

1. Show, using equations, how according to the Bronsted-Lowry theory the $H_2PO_4^-(aq)$ ion is amphiprotic/amphoteric. Identify the conjugate acid-base pairs in each reaction. Make sure to show all your work. (4 marks)



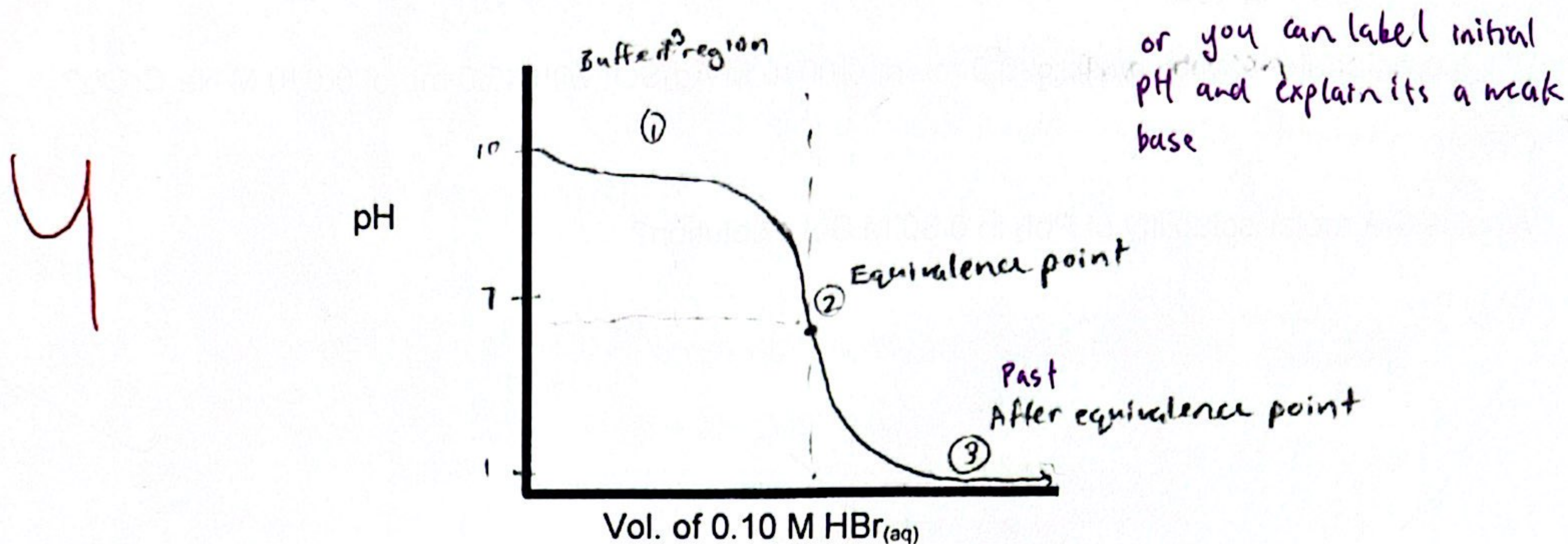
$H_2PO_4^-$ is able to act as both a proton donor and a proton acceptor, making it amphoteric



weak base + strong acid

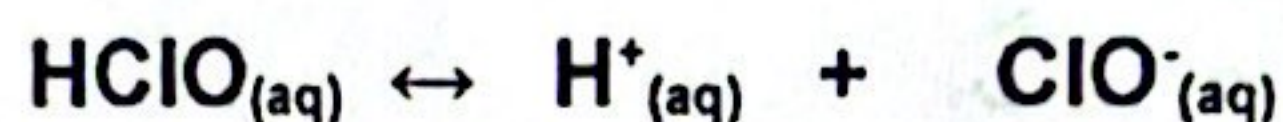
2. You are given a solution of $\text{NH}_3(\text{aq})$ that has a concentration of 0.10 M. You titrate the solution with 0.10 M $\text{HBr}(\text{aq})$. Sketch a detailed titration curve for the titration in the axis provided below, showing the equivalence point and its pH relative to pH=7. Label the names of the 3 distinct regions. (4 marks)

Titration of 0.10 M $\text{NH}_3(\text{aq})$ with $\text{HBr}(\text{aq})$



APPLICATION (9 MARKS)

1. Chlorine-containing compounds such as calcium hypochlorite are added to backyard swimming pools to kill the microorganisms in pool water and oxidize organic matter in the pool. Once dissolved in water the hypochlorite ion forms the equilibrium:



Maintaining the backyard pool at pH 7.4 – the pH of human tears – is ideal. At this level, the concentration of hypochlorous acid and hypochlorite are approximately equal.

a) Predict the effect that increasing pH has on the concentration of $\text{HClO}(\text{aq})$ in the pool, and explain in point form. (2 marks)

- increasing pH means lowering the concentration of H^+ ions
- concentration of products (H^+) is lowered
- equilibrium would shift right to make up for lost H^+ concentration

✓ The shift to the right would decrease the concentration of $\text{HClO}(\text{aq})$ (reactant)

b) Predict the effect of decreasing pH on the concentration of $\text{ClO}^-(\text{aq})$ and explain in point form. (2 marks)

- decreasing pH = increase in H^+ ion concentrations
- concentration of products (H^+) is increased
- equilibrium would shift left to accommodate for excess of products (H^+) concentration
- The shift left would decrease the concentration of $\text{ClO}^-(\text{aq})$ (product)

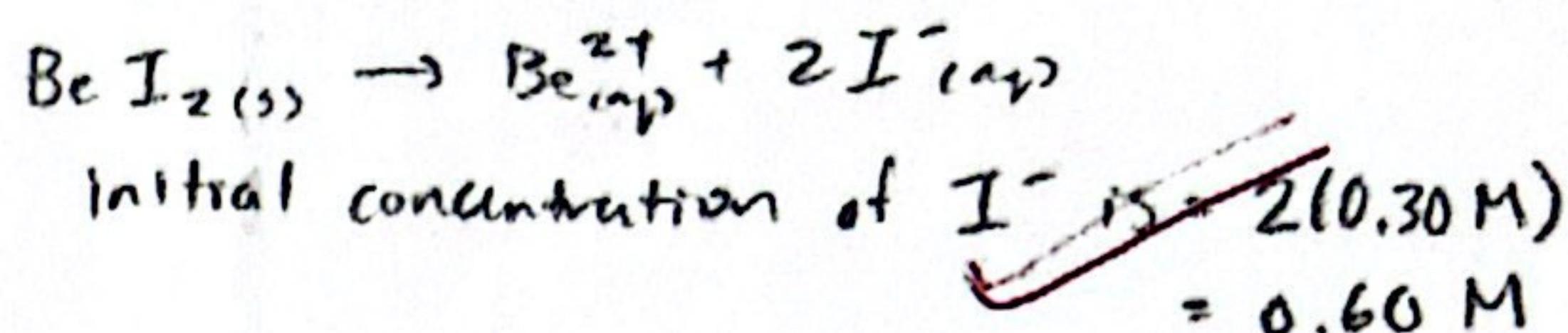
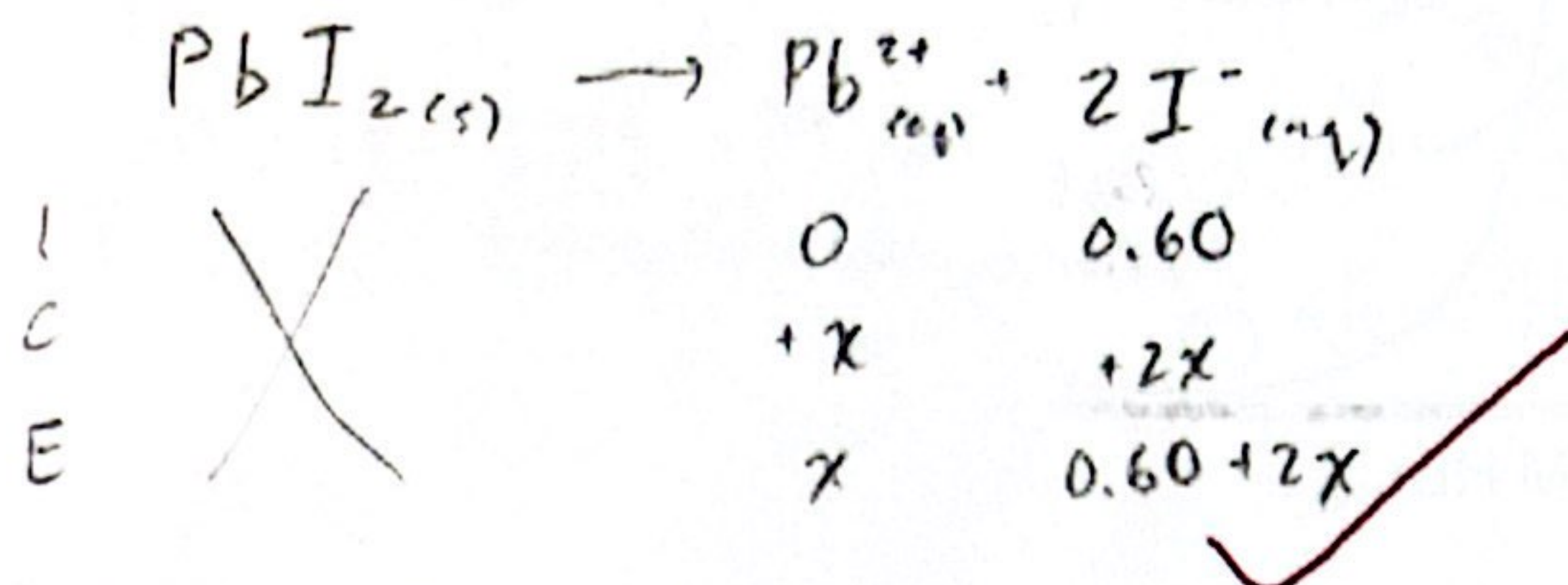
2. Choose 1 of the following to complete (5 marks):

Will a precipitate form by mixing 20.0 mL of 0.0010 M Ag_2SO_4 with 45.0 mL of 0.010 M Na_2CrO_4 ?

OR

i think the ans is yes $Q = 6.55 \times 10^{-11} \text{ M}$

What is the molar solubility of PbI_2 in 0.30 M BeI_2 solution?



$$K_{sp} = [\text{I}^-]^2 [\text{Pb}^{2+}]$$

$$= (0.60 + 2x)^2 (x)$$

$$8.5 \times 10^{-9} = (0.60)^2 (x)$$

$$\frac{8.5 \times 10^{-9}}{(0.60)^2} = x$$

$$2.36 \times 10^{-8} = x = [\text{Pb}^{2+}]$$

$$K_{sp} \text{ of } \text{PbI}_2 = 8.5 \times 10^{-9}$$

$$\frac{0.60 \text{ M}}{K_{sp}} = \frac{0.60 \text{ M}}{8.5 \times 10^{-9}} = 70588235.29 > 100$$

100 rule applies!

Molar ratio of $\text{Pb}^{2+}_{(aq)}$ to $\text{PbI}_2(s)$ is 1:1

$$\text{Molar solubility of } \text{PbI}_2(s) = 2.4 \times 10^{-8} \text{ mol/L}$$